

12_Py

April 25, 2025

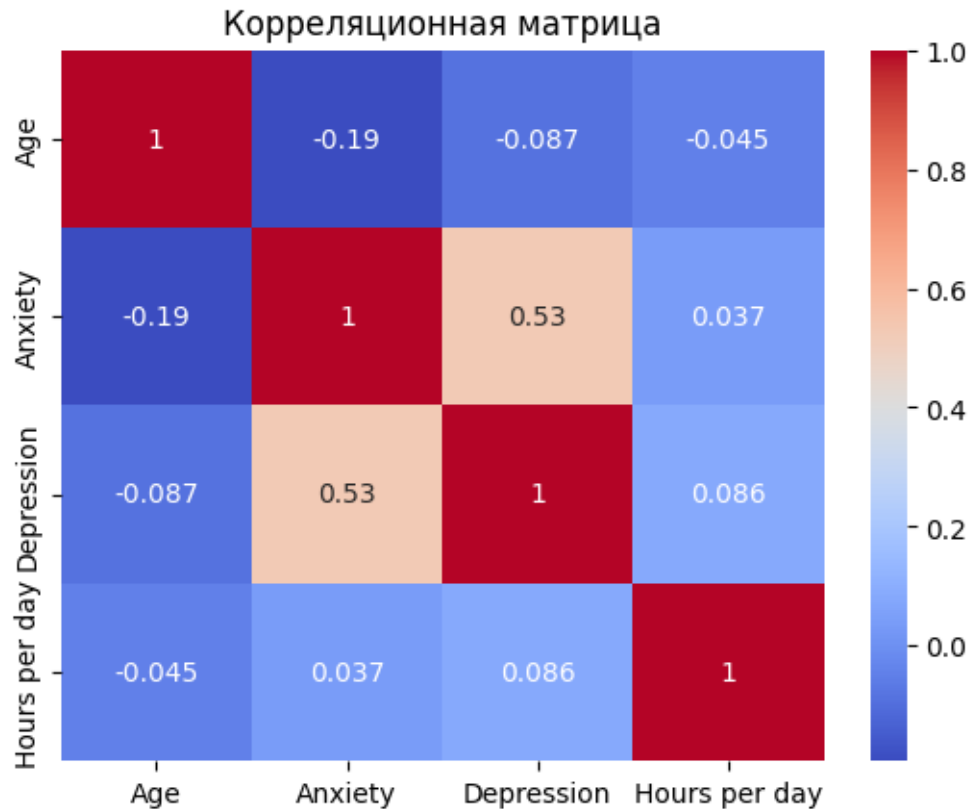
```
[1]: #      1 - EDA      Python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("mxmh_survey_results.csv")
df.dropna(inplace=True)

#
print(df[["Age", "Anxiety", "Depression", "Hours per day"]].describe())

#
corr_matrix = df[["Age", "Anxiety", "Depression", "Hours per day"]].corr()
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title(" ")
plt.show()
```

	Age	Anxiety	Depression	Hours per day
count	616.000000	616.000000	616.000000	616.000000
mean	24.792208	5.883929	4.894481	3.702435
std	11.658515	2.758269	2.998005	3.071961
min	10.000000	0.000000	0.000000	0.000000
25%	18.000000	4.000000	2.000000	2.000000
50%	21.000000	6.000000	5.000000	3.000000
75%	27.000000	8.000000	7.000000	5.000000
max	89.000000	10.000000	10.000000	24.000000



```
[2]: #      3 - Python
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

X = df[["Depression", "Hours per day", "Age"]]
y = df["Anxiety"]

model = LinearRegression().fit(X, y)
print(f"R²: {model.score(X, y):.2f}")
print(f"      : {model.coef_}")
```

R²: 0.30
 : [0.47482376 -0.01249656 -0.03532374]

```
[3]: #      5 - Python
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report

X = pd.get_dummies(df[["Age", "Anxiety", "Depression"]])
y = df["Music effects"]
```

```
model = LogisticRegression(max_iter=1000).fit(X, y)
print(classification_report(y, model.predict(X)))
```

	precision	recall	f1-score	support
Improve	0.76	1.00	0.86	465
No effect	1.00	0.01	0.01	136
Worsen	0.00	0.00	0.00	15
accuracy			0.76	616
macro avg	0.59	0.34	0.29	616
weighted avg	0.79	0.76	0.65	616

```
c:\Users\borga\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\metrics\_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.
```

```
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
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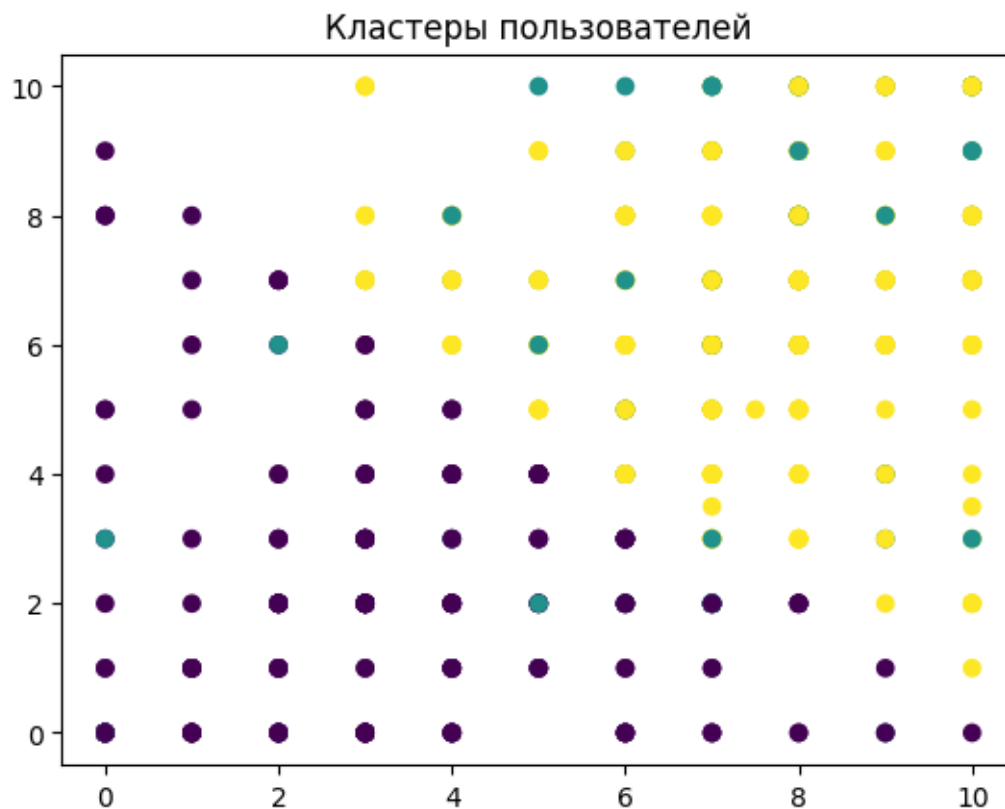
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```

```
[4]: # 7 - K-means
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

X = df[["Anxiety", "Depression", "Hours per day"]]
kmeans = KMeans(n_clusters=3).fit(X)
df["Cluster"] = kmeans.labels_

#
plt.scatter(df["Anxiety"], df["Depression"], c=df["Cluster"], cmap="viridis")
plt.title(" ")
plt.show()
```



```
[5]: !jupyter nbconvert --to pdf 12_Py.ipynb
```

```
[NbConvertApp] Converting notebook 12_Py.ipynb to pdf
[NbConvertApp] Support files will be in 12_Py_files\
[NbConvertApp] Making directory .\ 12_Py_files
[NbConvertApp] Writing 37350 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | b had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 86183 bytes to 12_Py.pdf
```